

Muhammad Fahad

+92-3329296026 - [Portfolio](#) - [Email](#) - [Linkedin](#) - [Github](#)

SUMMARY

I'm a **Full Stack Software Engineer** building practical solutions that make everyday tasks easier. I've worked with the **MERN**, **MEAN stack** and **Django** to create tools like a **smart university transport system** for my campus and a **productivity app for people with short-term memory** challenges, aligning my experience with office-based collaboration and deployment needs. I focus on writing clean, scalable code and building user-friendly experiences that solve real problems.

WORK EXPERIENCE

Junior Software Engineer *AitchTech Bahawalpur*

July 2025 - present

Magnt – Enterprise AI Chat Application

- Built **6 interactive chart types** (line, bar, pie, donut, graph, dot) using **Vega Charts**
- Engineered a **3-theme system** (light, dark, Magnt-branded) from scratch.
- Delivered **bilingual LTR/RTL layout** support for English and Arabic.

VaultX – Blockchain Based NFT Platform

- Integrated **Privy Wallet** to enable secure wallet connection and user authentication.
- Implemented **transaction signing** and **on-chain submission** for secure blockchain data storage.
- Connected frontend to **backend REST APIs** to ensure dynamic, real-time data handling.
- Replaced **Zustand state management** with **SWR**, improving data fetching, caching, and revalidation for a more reactive UI.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Frontend Technologies: Angular JS, React JS

Backend Technologies: Nest.js, Node.js, Django, Express.js, REST APIs

Databases: MongoDB, PostgreSQL, Redis

Cloud Computing: AWS-Basics (IAM, AMI, EC2, S3, RDS, DynamoDB, Route53, Lambda, CloudFront, CloudFormation, Load Balancer, ASG)

Deployment Platforms: Vercel, Netlify, Render, Digital Ocean

Version Control: Git & GitHub

DevOps & CI/CD: Docker, GitHub Actions, Automated CI/CD pipelines

Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn

PERSONAL PROJECTS

Smart University Bus Information System (FYP) | [GitHub](#)

- **Digitized** a university's paper-based transportation schedule into a live web platform with an **admin panel** for route/stop management and **telemetry logging**.
- Architected a **full-stack distributed system** with two independent microservices (**NestJS/PostgreSQL** for real-time tracking, **Node.js/MongoDB** for timetable management) and a **React** frontend.
- Implemented **real-time bus tracking** with a custom **user clustering algorithm** for crowd estimation at stops, plus automated student alerts and **ETA calculations**.
- Built **Redis caching** for **OTP verification** and **timetable data**, reducing database queries across both services
- Containerized all services with **Docker** and authored separate **Compose files** for production (bus-tracking only) and local development (full stack).
- Set up a **GitHub Actions** pipeline that triggers only on bus-tracking service changes, builds the **Docker image**, and redeploys to **AWS EC2** automatically.
- Deployed across **hybrid cloud infrastructure** bus-tracking on **AWS EC2**, timetable and frontend on **Vercel**.

Task AI Studio (MERN) | [GitHub](#)

- Built a **productivity app** to help users with **short-term memory** to track tasks and receive timely reminders.

- Implemented **AI-powered Task Scheduling** that automatically sorts tasks based on priority and deadlines to optimize user productivity.
 - Developed a Node.js-based **WhatsApp bot** that sends task **reminders every 3 hours**.
 - Enabled **WhatsApp-based task retrieval** using `whatsapp-web.js` library.
- Ecommerce Website (MERN) | [GitHub](#)*
- Converted a **WordPress** website into a high-performance **MERN stack** application.
 - Improved load time from **57s to 12s** (78% speed boost).
 - Built an **admin panel** with order, product, and user management.
 - Integrated **analytics** to track visitor engagement and product views.

EDUCATION

The Islamia University of Bahawalpur <i>BS in Computer Science, GPA: 3.4</i>	Bahawalpur, Pakistan <i>Aug 2022 - Present</i>
• Awarded an Appreciation Certificate for serving as an Official Member of the IUB Publication Society as a Designer/Editor. • Received a Shield in recognition of contribution to BLCF (Bahawalpur Literary Cultural Festival) organization.	
Military College Sui <i>F.Sc (Grades: 95%)</i>	Sui, Pakistan <i>May 2016 – Jul 2021</i>
• Awarded an Appreciation Certificate for being part of the Drama Club for 1 year. • Achieved 3rd Position in Inter House Painting Calligraphy Competition. • Actively participated in sports and physical activities including Volleyball and Running.	

CERTIFICATIONS

Responsive Web Design , FreeCodeCamp	Sep 2023
Intro to Git and GitHub , Google/Coursera	Dec 2023
React Basics , Meta/Coursera	Jan 2024
Intro To Python , DataCamp	Jan 2025
Writing Efficient Code in Python , DataCamp	Jan 2025
Exploratory Data Analysis , DataCamp	Jan 2025